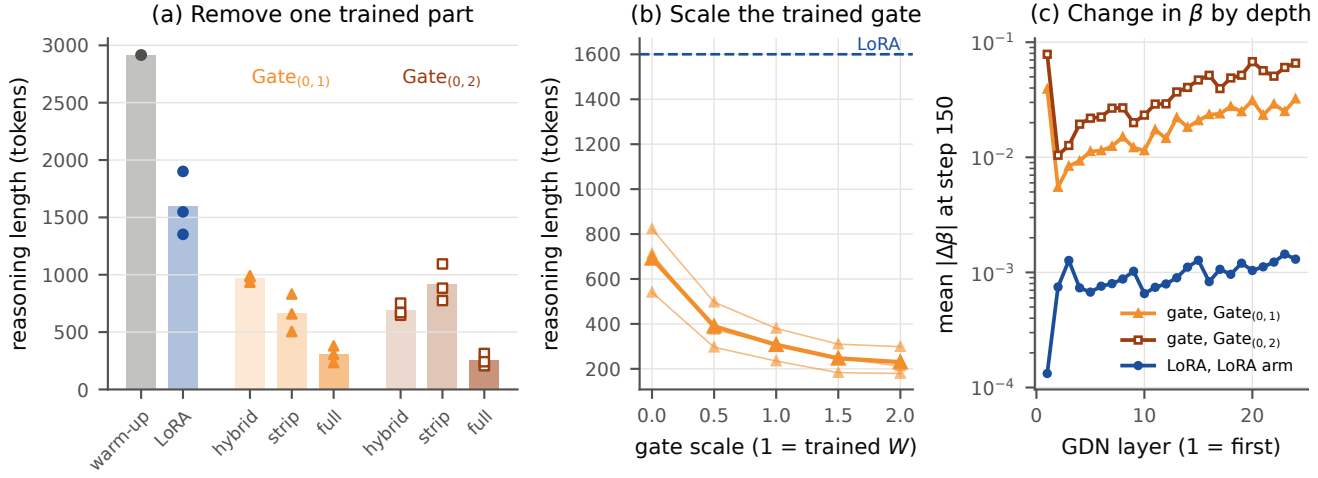
(a) Remove one trained part
reasoning length (tokens)
3000
2500
2000
1500
1000
500
0
Gate(0, 1)
Gate(0, 2)
warm-up
LoRA
hybrid
strip
full
hybrid
strip
full
(b) Scale the trained gate
reasoning length (tokens)
1600
1400
1200
1000
800
600
400
200
LoRA
0.0
0.5
1.0
1.5
2.0
gate scale (1 = trained W)
(c) Change in β by depth
mean |Δβ| at step 150
10−1
10−2
10−3
10−4
gate, Gate(0, 1)
gate, Gate(0, 2)
LoRA, LoRA arm
0
10
20
GDN layer (1 = first)